

| Assessment | Test | Domain                    | Skill                            | Difficulty                               |
|------------|------|---------------------------|----------------------------------|------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Right triangles and trigonometry | Hard <div></div> <div></div> <div></div> |

Question

$RS = 20$

$ST = 48$

$TR = 52$

The side lengths of right triangle  $RST$  are given. Triangle  $RST$  is similar to triangle  $UVW$ , where  $S$  corresponds to  $V$  and  $T$  corresponds to  $W$ . What is the value of  $\tan W$ ?

Answer

- A.  $\frac{5}{13}$
- B.  $\frac{5}{12}$
- C.  $\frac{12}{13}$
- D.  $\frac{12}{5}$

Correct Answer: B

Rationale

Choice B is correct. It's given that right triangle  $RST$  is similar to triangle  $UVW$ , where  $S$  corresponds to  $V$  and  $T$  corresponds to  $W$ . It's given that the side lengths of the right triangle  $RST$  are  $RS = 20$ ,  $ST = 48$ , and  $TR = 52$ . Corresponding angles in similar triangles are equal. It follows that the measure of angle  $T$  is equal to the measure of angle  $W$ . The hypotenuse of a right triangle is the longest side. It follows that the hypotenuse of triangle  $RST$  is side  $TR$ . The hypotenuse of a right triangle is the side opposite the right angle. Therefore, angle  $S$  is a right angle. The adjacent side of an acute angle in a right triangle is the side closest to the angle that is not the hypotenuse. It follows that the adjacent side of angle  $T$  is side  $ST$ . The opposite side of an acute angle in a right triangle is the side across from the acute angle. It follows that the opposite side of angle  $T$  is side  $RS$ . The tangent of an acute angle in a right triangle is the ratio of the length of the opposite side to the length of the adjacent side. Therefore,  $\tan T = \frac{RS}{ST}$ . Substituting  $20$  for  $RS$  and  $48$  for  $ST$  in this equation yields  $\tan T = \frac{20}{48}$ , or  $\tan T = \frac{5}{12}$ . The tangents of two acute angles with equal measures are equal. Since the measure of angle  $T$  is equal to the measure of angle  $W$ , it follows that  $\tan T = \tan W$ . Substituting  $\frac{5}{12}$  for  $\tan T$  in this equation yields  $\frac{5}{12} = \tan W$ . Therefore, the value of  $\tan W$  is  $\frac{5}{12}$ .

Choice A is incorrect. This is the value of  $\sin W$ .

Choice C is incorrect. This is the value of  $\cos W$ .

Choice D is incorrect. This is the value of  $\frac{1}{\tan W}$ .